LexKeyPlan: Planning with Keyphrases and Retrieval Augmentation for Legal Text Generation: A Case Study on European Court of Human Rights Cases

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Abstract

Large language models excel at legal text generation but often produce hallucinations due to their sole reliance on parametric knowledge. Retrieval-augmented models mitigate this by providing relevant external documents to the model but struggle when retrieval is based only on past context, which may not align with the model's intended future content. We introduce LexKeyPlan, a novel framework that integrates anticipatory planning into generation. Instead of relying solely on context for retrieval, LexKeyPlan generates keyphrases outlining future content serving as forward-looking plan, guiding retrieval for more accurate text generation. This work incorporates planning into legal text generation, demonstrating how keyphrases—representing legal concepts—enhance factual accuracy. By structuring retrieval around legal concepts, LexKeyPlan better aligns with legal reasoning, making it particularly suited for legal applications. Using the ECHR corpus as case study, we show that LexKeyPlan improves factual accuracy and coherence by retrieving information aligned with the intended content.

1 Introduction

Recent advancements in large language models (LLMs) have led to their adoption in the legal domain for tasks such as sifting through case briefs, expediting legal research, drafting contracts, and formulating litigation strategies (Dahl et al., 2024). Despite their ability to pass bar and law school exams (Katz et al., 2024; Martínez, 2024), perform statutory reasoning and interpretation (Blair-Stanek et al., 2023; Engel and Mcadams, 2024), and apply legal reasoning frameworks like Issue-Rule-Application-Conclusion (IRAC) (Kang et al., 2023; Guha et al., 2024), LLMs still struggle with factual inaccuracies and hallucinations of legal knowledge.

Traditional LLMs generate responses based on the input context and their parametric knowledge (Radford et al., 2019; Zhang et al., 2022; Jiang et al., 2023; Touvron et al., 2023), but this approach often fails to ensure factual accuracy. Retrieval-augmented generation (RAG) addresses this issue by retrieving external documents based on the context, allowing the model to condition its responses on more reliable information (Lewis et al., 2020; Izacard et al., 2023; Borgeaud et al., 2022; Ram et al., 2023). However, RAG relies entirely on the retrieval mechanism's ability to fetch relevant knowledge, which may not always align with the content the model intends to generate. This limitation is particularly critical in legal text generation, where context alone may not provide sufficient signals to retrieve legally relevant information (Magesh et al., 2024; Santosh et al., 2025).

To overcome this challenge, we propose LexKeyPlan, a novel framework that introduces an anticipatory planning stage into the generation process. Instead of relying solely on context for retrieval, LexKeyPlan first generates keyphrases that represent legal concepts outlining the intended content of the response. These keyphrases serve as a forward-looking content plan, guiding the retrieval of relevant documents from external sources. By shifting the retrieval process from purely reactive to proactively structured around anticipated content, LexKeyPlan better aligns retrieval with legal reasoning, reduces reliance on misleading contextual information, and improves coherence in long-form legal text generation.

This work incorporates content planning into legal text generation, highlighting how keyphrases—representing legal concepts—can enhance retrieval, factual grounding, and structured reasoning. Using the European Court of Human Rights (ECHR) case law corpus, we demonstrate that LexKeyPlan significantly improves legal text generation by reducing hallucinations and enhancing retrieval relevance. Our findings suggest that anticipatory content planning through legal concepts or keyphrases is crucial for legal applications, where precise retrieval and structured reasoning are essential for generating legally sound text.

2 LexKeyPlan

We propose a three-step framework for legal text generation, LexKeyPlan. Unlike traditional models that generate responses directly from context, LexKeyPlan introduces an intermediate planning stage using keyphrases. These keyphrases serve as a forward-looking blueprint, outlining the intent of the future response. By anticipating the necessary content, they guide the retrieval of relevant external documents, ensuring that the generated text is both factually accurate and grounded. This approach contrasts with existing retrieval augmentation methods, which typically use context alone to retrieve relevant information, potentially limiting the relevance of the retrieved content for response generation, in long form text generation.

Formally, let x be the input context. The model first generates a content plan c for the response, which can be represented as $p(c \mid x)$. This content plan, consisting of keyphrases, succinctly captures the essence of the future response. Then the generated plan c is used as query to retrieve relevant documents d from external knowledge sources. The final response y is produced by conditioning the model on the input context, the content plan and the retrieved documents, represented as $p(y \mid c, p, d)$.

Training the Content Plan Generation Module: We train a language model using the input context followed by the content plan. To obtain the target content plan for each input, we utilize the target response text and extract keyphrases from it. We explore two keyphrase extraction algorithms to create the supervision dataset for training this content plan generation module: TextRank (Mihalcea and Tarau, 2004), a graph-based ranking algorithm inspired by PageRank, and KeyBERT (Grootendorst, 2020), leveraging embeddings from pre-trained models.

Retriever Module: We investigate two off-the-shelf retrievers: 1) BM25 (Robertson et al., 2004), a sparse lexical-based model, and 2) GTR (Ni et al., 2021), an embedding-based dense retriever model built on T5-XXL (Raffel et al., 2020). We provide the concatenated list of keyphrases (content plan) as query to retrieve the relevant documents from datastore, built from ECHR judgement documents.

Training the Response Generation Module: The language model is trained using the input context, content plan, retrieved relevant documents followed by the target response. Since we do not have a gold-standard content plan or relevant documents for training, we extract keyphrases from the target response to serve as the content plan and use the target response as a query to retrieve pseudo-golden relevant documents. During inference, the model uses the actual generated content plan to retrieve relevant documents.

This three-step framework offers controllability, enabling users to adjust the content plan or retrieved documents to steer the model's response.

3 Experiments

3.1 Dataset & Metrics

ECHR CaseLaw consists of case judgments heard by the European Court of Human Rights. We use the latest cleaned version of this dataset from Santosh et al. (2024), which contains 15,729 cases in English from 1960 to July 28, 2022 and use the provided chronological splits for training (1960-2016), validation (2017-2018) and test set (2018-2022). Each case is segmented into two main sections: *The Facts*, which details the factual background of each case and *The Law*, which presents the legal reasoning justifying the outcome concerning alleged violations of specific ECHR articles. Our task involves generating the *The Law* (reasoning) section based on the *The Facts* section. While generating an entire section would be ideal, evaluating such large outputs against reference texts poses significant challenges. Therefore, we focus on paragraph-level generation to facilitate more accurate evaluations. Specifically, each paragraph in the *The Law* section serves as the target response, while the input context consists of the *The Facts* section and all preceding paragraphs from *The Law* section up to the target paragraph. During training, each paragraph is appended with an $< |end\_of\_paragraph| >$ token, allowing us to generate the next paragraph up to this marker during open-ended generation. We use `Mistral-7B-Instruct-v0.1` (Jiang et al., 2023) as our base LM for both plan and response generator and follow parameter efficient fine-tuning using LoRA (Hu et al., 2021). Implementation details are provided in App. B.

We evaluate the quality of the generated paragraphs: ROUGE-1,2,L (Lin, 2004) for lexical overlap with the reference paragraph, BERTScore (Zhang et al., 2019) for semantic similarity between generated and reference paragraph, Align-

	Plan Generator	Retriever	R-1 / R-2 / R-L	BERT Score	Align Score	Coh. / Flu.
a	-	-	0.32 / 0.13 / 0.21	0.77	0.52	0.69 / 0.61
b	TextRank	-	0.33 / 0.13 / 0.22	0.72	0.54	0.74 / **0.66**
c	KeyBERT	-	0.34 / 0.14 / 0.21	0.74	0.58	0.72 / 0.64
d	-	BM25	0.37 / 0.14 / 0.23	0.78	0.64	0.63 / 0.61
e	-	GTR	0.37 / 0.15 / 0.24	0.79	0.66	0.64 / 0.63
f	TextRank	BM25	0.37 / **0.17** / **0.26**	0.79	0.7	0.72 / 0.65
g	TextRank	GTR	0.39 / 0.15 / 0.24	0.78	0.72	0.75 / 0.65
h	KeyBERT	BM25	**0.40** / 0.15 / 0.23	0.78	0.73	0.74 / **0.66**
i	KeyBERT	GTR	0.39 / 0.16 / 0.24	**0.80**	**0.75**	**0.76** / 0.65

Table 1: Comparison of LexKeyPlan with Baseline Approaches. The "Plan Generator" column specifies the keyphrase extraction algorithm used to generate supervision signals for training both the planning and response generation modules. The "Retriever" column indicates the retrieval method employed during inference and for obtaining relevant documents during training of the response generator. Note that the keyphrase-based content plan is automatically generated by the model during inference.

Score (Zha et al., 2023) for factual consistency based on a unified alignment function between the reference and generated text and UniEval (Zhong et al., 2022) that evaluates coherence and fluency of the generated paragraph with respect to the context.

3.2 Results

To demonstrate the effectiveness of LexKeyPlan, we compare our method with several baselines: (a) traditional fine-tuning without keyphrase-based content planning or retrieval augmentation; (b, c) models with keyphrase-based planning, followed by response generation using the context and the generated plan. Two keyphrase extraction algorithms are used as supervision signals for training the planning and response generators; (d, e) models without keyphrase planning that directly retrieve relevant documents based on context and generate responses using the retrieved documents. The response generation models are trained with supervision from their respective retrievers; (f, g, h, i) our three-staged LexKeyPlan method, with different combinations of plan generators and retrievers for training the response generation.

Effect of Keyphrase-based Content Planning: As seen in Table 1, keyphrase-based content planning (b, c) outperforms traditional fine-tuning (a) in terms of coherence, fluency, and faithfulness (AlignScore), while remaining comparable on lexical similarity (ROUGE). This keyphrase planning mechanism allows the model to anticipate and structure its response around key concepts in the upcoming text. By separating the planning and generation stages, the model's cognitive load is reduced, as it no longer needs to generate relevant information and maintain coherence simultaneously. Instead, it can focus on each task independently, resulting in more coherent and contextually accurate outputs. Additionally, this structured planning mitigates hallucinations by providing a clear blueprint for the text, helping the model stay grounded in the key content and reducing the likelihood of producing irrelevant information. Between the two keyphrase algorithms, KeyBERT outperforms on content-related metrics, while TextRank excels in stylistic aspects like coherence and fluency.

Effect of Retrieval Augmentation: Retrieval augmentation (d, e), which utilizes context as the query, effectively addresses hallucination issues by grounding the generated text in external knowledge, resulting in improved ROUGE scores and faithfulness (AlignScore). However, this approach also leads to a decrease in coherence and fluency, likely due to the model's challenge in integrating necessary information from the text while maintaining coherent output. Among the retrieval methods, embedding-based GTR consistently outperforms BM25 across all metrics, revealing the limitations of lexical-based retrievers in capturing semantic nuances. Nonetheless, the smaller gap for BM25 highlights the high lexical overlap and repetitive nature of legal concepts in the documents (Santosh et al., 2024), making BM25 a scalable and efficient option for legal text retrieval.

LexKeyPlan: Integrating keyphrase-based content planning with retrieval augmentation significantly improves both content quality and stylistic metrics (statistically significant improvements as per Wilcoxon signed-rank test at a 95% confidence interval over the baselines without planning or retrieval augmentation). By using keyphrases as queries to anticipate forthcoming text, LexKeyPlan offers two key advantages. First, it enhances retrieval accuracy by focusing on key concepts, lead-

	Plan	Retreiver	R-1 / R-2 / R-L	BERT Score	Align Score	Coh. / Flu.
a	-	-	0.23 / 0.06 / 0.15	0.72	0.44	0.70 / 0.64
b	-	BM25	0.25 / 0.08 / 0.15	0.74	0.56	0.60 / 0.62
c	-	GTR	0.25 / 0.08 / 0.17	0.75	0.58	0.62 / 0.62
d	✓	-	0.24 / 0.08 / 0.16	0.74	0.50	0.72 / 0.62
e	✓	BM25	0.26 / 0.10 / 0.18	0.76	0.59	0.58 / 0.61
f	✓	GTR	0.27 / 0.10 / 0.16	0.75	0.58	0.59 / 0.60

Table 2: Effect of Integrating Keyphrase-Based Content Planning and Retrieval Augmentation in Zero-Shot Setting. The 'Plan' column specifies whether the model is prompted to generate keyphrase-based content plan for the next response. The Retriever column identifies the retrieval method employed.

	Similarity	Exact Match
Zeroshot	0.67	0.18
Fine-tuned with TextRank	0.72	0.36
Fine-tuned with KeyBERT	0.78	0.42
TextRank (Oracle)	0.96	1.0
KeyBERT (Oracle)	0.95	1.0

Table 3: Evaluation of Content Plan Quality.

ing to more relevant documents compared to using context alone, which may miss crucial information for the anticipated text. Second, separating the planning and generation tasks allows the model to produce more coherent text. Rather than performing retrieval, integration, and generation simultaneously, LexKeyPlan enables the model to first plan around keyphrases, thereby reducing cognitive load and resulting in more coherent and contextually accurate outputs. Among the methods evaluated, KeyBERT provided superior supervision for plan generation, while GTR outperformed BM25, highlighting the benefits of embedding-based models over traditional lexical approaches.

Zero-shot Experiments: To evaluate whether integrating content planning and retrieval augmentation into pre-trained models in a zero-shot setting enhances performance, we compare several approaches in Table 2: (a) prompting the pre-trained model to generate a response based solely on the context; (b, c) prompting the model to generate a response based on both the context and relevant documents retrieved from the context; (d) prompting the model to first create a content plan in the form of keyphrases and then generate a response based on this plan; (e, f) prompting the model to generate a content plan, use this plan to retrieve relevant documents, and then generate a response based on both the plan and documents.

As expected, retrieval augmentation (b, c) directly enhances faithfulness (AlignScore) and content quality but results in a decrease in coherence. This drop underscores the difficulty of maintaining coherence while incorporating relevant information. Introducing content planning (d) helps the model manage these tasks by breaking them into sequential steps, which enhances both faithfulness and coherence. When combining content planning with retrieval augmentation (e, f), there is a slight improvement in content quality metrics, though stylistic scores decrease. This suggests that zero-shot models may face challenges with the complexity of handling multiple instructions—such as following a content plan and integrating relevant information.

Analysis of Keyphrases in Content Planning: We evaluate the quality of keyphrases generated through (a) Zero-shot prompting the pre-trained model, and (b, c) using a fine-tuned model supervised with TextRank and KeyBERT, by computing the embedding similarity of each keyphrase to the actual target (i.e., the next paragraph). We report both average similarity and exact match (if the keyphrase appears in the target). Additionally, we present oracle metrics (d, e), where keyphrases are directly extracted from the target paragraph using TextRank and KeyBERT. Results show that fine-tuning with KeyBERT-extracted keyphrases leads to more relevant keyphrase generation with respect to future content, providing informed plans for document retrieval and response generation.

4 Conclusion

We introduce LexKeyPlan, a three-step framework for legal text generation integrating keyphrase-based content planning with retrieval augmentation. Unlike traditional methods that rely solely on context for retrieval and generation, LexKeyPlan first generates a content plan outlining the intended future content. This plan not only guides to produce coherent response but also the retrieval of relevant documents. LexKeyPlan through its anticipatory planning ensures that the generated text is both contextually coherent and factually accurate. as demonstrated with the ECHR case law corpus.

Limitations

LexKeyPlan currently utilizes a simple content planning approach through keyphrases, guided by algorithms like TextRank and KeyBERT. These algorithms, though effective, were primarily developed for general corpora like news articles, scientific papers, and other non-legal texts. They are designed to extract keyphrases that are broadly applicable but may not fully address the unique requirements and nuances of legal language and context. Future research could explore advanced keyphrase extraction methods specifically tailored to legal texts, such as Mandal et al. (2017). Such methods could handle the distinct terminology and complex structures characteristic of legal language, improving the relevance and precision of keyphrases used in content planning.

More sophisticated content planning approaches could significantly enhance the effectiveness of LexKeyPlan. One promising avenue is the use of graphical representations derived from legal concept networks. These networks visually map out legal concepts and their interrelationships, creating a structured framework that can guide content generation. Incorporating a specialized legal thesaurus into the planning process can further refine content generation. A legal thesaurus provides a curated vocabulary of legal terms and concepts, enhancing the precision of keyphrases and improving the contextual relevance of the generated content. Improving the integration of content plans during the text generation phase may also involve adopting constrained decoding techniques. Constrained decoding methods enforce adherence to predefined content plans, ensuring that the generated text aligns closely with the planned structure and content. This approach can help maintain coherence and fidelity, reducing deviations from the intended content and improving output quality.

Currently, LexKeyPlan's performance is evaluated using established metrics like ROUGE, BERTScore, and AlignScore. While these metrics offer quantitative insights, they may not fully capture the complexities of legal content. Future research could develop domain-specific evaluation metrics tailored to the legal field. Another limitation of our study is the absence of direct validation by legal experts in the assessment of outputs, which we could not perform due to lack of access to legal experts. Lastly, while our evaluation was based on the ECHR corpus, future work should extend LexKeyPlan to handle legal texts across different languages and jurisdictions. Adapting keyphrase extraction and content planning processes to accommodate diverse legal terminologies and frameworks will enhance the system's applicability.

Ethics Statement

The ECHR dataset used in this study is publicly available and widely employed in legal NLP research. However, it is important to note that this dataset is not anonymized and contains real names and details of the parties involved in the cases. Although we do not foresee direct harm resulting from our experiments, handling such sensitive data necessitates careful consideration of privacy and ethical implications. Researchers must be diligent in protecting individual privacy and managing sensitive information responsibly, even when using publicly accessible datasets.

LLMs utilized in legal contexts have significant implications for both legal professionals and the public. Despite their advanced capabilities, these models carry inherent risks. They may reproduce and amplify biases inherent in their training data, leading to outputs that reflect historical inequalities or stereotypes. Additionally, LLMs may produce incorrect or misleading information, particularly in complex legal scenarios where precision is essential. Our research focuses on enhancing rather than replacing human expertise. It is crucial to apply LLM-generated outputs with careful scrutiny, ensuring that these tools are complemented by thorough human oversight and critical evaluation. Our aim is to improve legal text generation while recognizing that AI must be a supportive tool, used with an awareness of its limitations and potential impact on legal practice.

References

Griffin Adams, Alexander R Fabbri, Faisal Ladhak, Kathleen McKeown, and Noémie Elhadad. 2023. Generating edu extracts for plan-guided summary re-ranking. In *Proceedings of the conference. Association for Computational Linguistics. Meeting*, volume 2023, page 2680. NIH Public Access.

Regina Barzilay and Michael Elhadad. 1997. Using lexical chains for text summarization.

Regina Barzilay and Mirella Lapata. 2008. Modeling local coherence: An entity-based approach. *Computational Linguistics*, 34(1):1–34.

Andrew Blair-Stanek, Nils Holzenberger, and Benjamin Van Durme. 2023. Can gpt-3 perform statutory reasoning? In *Proceedings of the Nineteenth International Conference on Artificial Intelligence and Law*, pages 22–31.

Sebastian Borgeaud, Arthur Mensch, Jordan Hoffmann, Trevor Cai, Eliza Rutherford, Katie Millican, George Bm Van Den Driessche, Jean-Baptiste Lespiau, Bogdan Damoc, Aidan Clark, et al. 2022. Improving language models by retrieving from trillions of tokens. In *International conference on machine learning*, pages 2206–2240. PMLR.

Matthew Dahl, Varun Magesh, Mirac Suzgun, and Daniel E Ho. 2024. Large legal fictions: Profiling legal hallucinations in large language models journal of legal analysis (forthcoming).

Christoph Engel and Richard H Mcadams. 2024. Asking gpt for the ordinary meaning of statutory terms. *MPI Collective Goods Discussion Paper*, (2024/5).

Ameya Godbole, Nicholas Monath, Seungyeon Kim, Ankit Singh Rawat, Andrew Mccallum, and Manzil Zaheer. 2024. Analysis of plan-based retrieval for grounded text generation. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 13101–13119.

Maarten Grootendorst. 2020. Keybert: Minimal keyword extraction with bert.

Neel Guha, Julian Nyarko, Daniel Ho, Christopher Ré, Adam Chilton, Alex Chohlas-Wood, Austin Peters, Brandon Waldon, Daniel Rockmore, Diego Zambrano, et al. 2024. Legalbench: A collaboratively built benchmark for measuring legal reasoning in large language models. *Advances in Neural Information Processing Systems*, 36.

Markus Guhe. 2020. *Incremental conceptualization for language production*. Psychology Press.

Eduard H Hovy. 1993. Automated discourse generation using discourse structure relations. *Artificial intelligence*, 63(1-2):341–385.

Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2021. Lora: Low-rank adaptation of large language models. *arXiv preprint arXiv:2106.09685*.

Fantine Huot, Joshua Maynez, Shashi Narayan, Reinald Kim Amplayo, Kuzman Ganchev, Annie Priyadarshini Louis, Anders Sandholm, Dipanjan Das, and Mirella Lapata. 2023. Text-blueprint: An interactive platform for plan-based conditional generation. In *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics: System Demonstrations*, pages 105–116.

Gautier Izacard, Patrick Lewis, Maria Lomeli, Lucas Hosseini, Fabio Petroni, Timo Schick, Jane Dwivedi-Yu, Armand Joulin, Sebastian Riedel, and Edouard Grave. 2023. Atlas: Few-shot learning with retrieval augmented language models. *Journal of Machine Learning Research*, 24(251):1–43.

Albert Q Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, Lucile Saulnier, et al. 2023. Mistral 7b. *arXiv preprint arXiv:2310.06825*.

Xiaoxi Kang, Lizhen Qu, Lay-Ki Soon, Adnan Trakic, Terry Yue Zhuo, Patrick Charles Emerton, and Genevieve Grant. 2023. Can chatgpt perform reasoning using the irac method in analyzing legal scenarios like a lawyer? *arXiv preprint arXiv:2310.14880*.

Nikiforos Karamanis. 2004. Entity coherence for descriptive text structuring.

Daniel Martin Katz, Michael James Bommarito, Shang Gao, and Pablo Arredondo. 2024. Gpt-4 passes the bar exam. *Philosophical Transactions of the Royal Society A*, 382(2270):20230254.

Rodger Kibble and Richard Power. 2004. Optimizing referential coherence in text generation. *Computational Linguistics*, 30(4):401–416.

Diederik P Kingma and Jimmy Ba. 2014. Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980*.

Anirban Laha, Parag Jain, Abhijit Mishra, and Karthik Sankaranarayanan. 2020. Scalable micro-planned generation of discourse from structured data. *Computational Linguistics*, 45(4):737–763.

Willem JM Levelt. 1993. *Speaking: From intention to articulation*. MIT press.

Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, et al. 2020. Retrieval-augmented generation for knowledge-intensive nlp tasks. *Advances in Neural Information Processing Systems*, 33:9459–9474.

Chin-Yew Lin. 2004. Rouge: A package for automatic evaluation of summaries. In *Text summarization branches out*, pages 74–81.

Zhengyuan Liu and Nancy F Chen. 2021. Controllable neural dialogue summarization with personal named entity planning. *arXiv preprint arXiv:2109.13070*.

Varun Magesh, Faiz Surani, Matthew Dahl, Mirac Suzgun, Christopher D Manning, and Daniel E Ho. 2024. Hallucination-free? assessing the reliability of leading ai legal research tools. *arXiv preprint arXiv:2405.20362*.

Arpan Mandal, Kripabandhu Ghosh, Arindam Pal, and Saptarshi Ghosh. 2017. Automatic catchphrase identification from legal court case documents. In *Proceedings of the 2017 ACM on Conference on Information and Knowledge Management*, pages 2187–2190.

William C Mann and Sandra A Thompson. 1988. Rhetorical structure theory: Toward a functional theory of text organization. *Text-interdisciplinary Journal for the Study of Discourse*, 8(3):243–281.

Eric Martínez. 2024. Re-evaluating gpt-4's bar exam performance. *Artificial Intelligence and Law*, pages 1–24.

Kathleen McKeown and Dragomir R Radev. 1995. Generating summaries of multiple news articles. In *Proceedings of the 18th annual international ACM SIGIR conference on Research and development in information retrieval*, pages 74–82.

Kathleen R McKeown. 1985. Discourse strategies for generating natural-language text. *Artificial intelligence*, 27(1):1–41.

Chris Mellish, Alistair Knott, Jon Oberlander, and Mick O'Donnell. 1998. Experiments using stochastic search for text planning. In *Proceedings of the 9th International General Workshop*, pages 98–107. ACL Anthology.

Rada Mihalcea and Paul Tarau. 2004. Textrank: Bringing order into text. In *Proceedings of the 2004 conference on empirical methods in natural language processing*, pages 404–411.

Amit Moryossef, Yoav Goldberg, and Ido Dagan. 2019. Step-by-step: Separating planning from realization in neural data-to-text generation. *arXiv preprint arXiv:1904.03396*.

Shashi Narayan, Joshua Maynez, Jakub Adamek, Daniele Pighin, Blaz Bratanic, and Ryan McDonald. 2020. Stepwise extractive summarization and planning with structured transformers. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 4143–4159.

Shashi Narayan, Joshua Maynez, Reinald Kim Amplayo, Kuzman Ganchev, Annie Louis, Fantine Huot, Anders Sandholm, Dipanjan Das, and Mirella Lapata. 2023. Conditional generation with a question-answering blueprint. *Transactions of the Association for Computational Linguistics*, 11:974–996.

Shashi Narayan, Yao Zhao, Joshua Maynez, Gonçalo Simões, Vitaly Nikolaev, and Ryan McDonald. 2021. Planning with learned entity prompts for abstractive summarization. *Transactions of the Association for Computational Linguistics*, 9:1475–1492.

Jianmo Ni, Chen Qu, Jing Lu, Zhuyun Dai, Gustavo Hernández Ábrego, Ji Ma, Vincent Y Zhao, Yi Luan, Keith B Hall, Ming-Wei Chang, et al. 2021. Large dual encoders are generalizable retrievers. *arXiv preprint arXiv:2112.07899*.

Ratish Puduppully, Li Dong, and Mirella Lapata. 2019. Data-to-text generation with entity modeling. *arXiv preprint arXiv:1906.03221*.

Ratish Puduppully, Yao Fu, and Mirella Lapata. 2022. Data-to-text generation with variational sequential planning. *Transactions of the Association for Computational Linguistics*, 10:697–715.

Ratish Puduppully and Mirella Lapata. 2021. Data-to-text generation with macro planning. *Transactions of the Association for Computational Linguistics*, 9:510–527.

Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, Ilya Sutskever, et al. 2019. Language models are unsupervised multitask learners. *OpenAI blog*, 1(8):9.

Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J Liu. 2020. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of machine learning research*, 21(140):1–67.

Ori Ram, Yoav Levine, Itay Dalmedigos, Dor Muhlgay, Amnon Shashua, Kevin Leyton-Brown, and Yoav Shoham. 2023. In-context retrieval-augmented language models. *Transactions of the Association for Computational Linguistics*, 11:1316–1331.

Ehud Reiter and Robert Dale. 1997. Building applied natural language generation systems. *Natural Language Engineering*, 3(1):57–87.

Stephen Robertson, Hugo Zaragoza, and Michael Taylor. 2004. Simple bm25 extension to multiple weighted fields. In *Proceedings of the thirteenth ACM international conference on Information and knowledge management*, pages 42–49.

TYS Santosh, Rashid Gustav Haddad, and Matthias Grabmair. 2024. Ecthr-pcr: A dataset for precedent understanding and prior case retrieval in the european court of human rights. *arXiv preprint arXiv:2404.00596*.

TYSS Santosh, Isaac Misael OlguÃn Nolasco, and Matthias Grabmair. 2025. Lecopcr: Legal concept-guided prior case retrieval for european court of human rights cases. *arXiv preprint arXiv:2501.14114*.

Yijia Shao, Yucheng Jiang, Theodore Kanell, Peter Xu, Omar Khattab, and Monica Lam. 2024. Assisting in writing wikipedia-like articles from scratch with large language models. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 6252–6278.

Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, et al. 2023. Llama: Open and efficient foundation language models. *arXiv preprint arXiv:2302.13971*.

Yuheng Zha, Yichi Yang, Ruichen Li, and Zhiting Hu. 2023. Alignscore: Evaluating factual consistency with a unified alignment function. *arXiv preprint arXiv:2305.16739*.

Susan Zhang, Stephen Roller, Naman Goyal, Mikel Artetxe, Moya Chen, Shuohui Chen, Christopher Dewan, Mona Diab, Xian Li, Xi Victoria Lin, et al. 2022. Opt: Open pre-trained transformer language models. *arXiv preprint arXiv:2205.01068*.

Tianyi Zhang, Varsha Kishore, Felix Wu, Kilian Q Weinberger, and Yoav Artzi. 2019. Bertscore: Evaluating text generation with bert. *arXiv preprint arXiv:1904.09675*.

Ming Zhong, Yang Liu, Da Yin, Yuning Mao, Yizhu Jiao, Pengfei Liu, Chenguang Zhu, Heng Ji, and Jiawei Han. 2022. Towards a unified multi-dimensional evaluator for text generation. *arXiv preprint arXiv:2210.07197*.

Appendix

A Related Work

Content Planning for Text Generation: Structured planning is considered a critical link in organizing content effectively before realization (Reiter and Dale, 1997), much like humans plan at a higher level than individual words, as evidenced by psycholinguistic studies (Levelt, 1993; Guhe, 2020). Earlier approaches incorporated various planning representations, such as Rhetorical Structure Theory (Mann and Thompson, 1988; Hovy, 1993) and MUC-style representations (McKeown and Radev, 1995), discourse trees (Mellish et al., 1998), entity transitions (Kibble and Power, 2004; Barzilay and Lapata, 2008), sequences of propositions (Karamanis, 2004), schemas (McKeown, 1985) and lexical chains (Barzilay and Elhadad, 1997).

Recent works in the data-to-text generation task divide it into two distinct phases: planning and realization of natural language text (Moryossef et al., 2019). Puduppully et al. (2019) and Laha et al. (2020) proposed micro-planning strategies, where they first establish a content plan based on a sequence of records and then generate a summary conditioned on that plan. Similar planning approaches have been explored for entity realization (Puduppully et al., 2019). Additionally, Puduppully and Lapata (2021) advocated for macro-planning as a means of organizing high-level document content, either in text form or within latent space (Puduppully et al., 2022).

In summarization, various plan representations have been investigated. Narayan et al. (2020) treat step-by-step content selection as a planning component, generating sentence-level plans through extract-then-abstract methods, and even sub-sentence-level plans using elementary discourse units (Adams et al., 2023). Further research by Narayan et al. (2021); Liu and Chen (2021) introduced intermediate plans using entity chains—ordered sequences of entities mentioned in the summary. More recently, Narayan et al. (2023); Huot et al. (2023) conceptualized text plans as a sequence of question-answer pairs, which serve as blueprints for content selection (i.e., what to say) and planning (i.e., in what order) in summarization tasks.

Recently, Godbole et al. (2024); Shao et al. (2024) have explored planning-based approaches for text generation in tasks such as expository writing and biographical summaries. However, legal text generation presents unique challenges that remain unexplored in this context. Unlike traditional QA tasks, which involve focused information-seeking queries, legal text generation requires reasoning over long, complex case facts to construct well-grounded arguments to derive conclusions. This makes planning particularly valuable, as it can help structure retrieval and generation to ensure coherence, factual consistency, and legal validity. To the best of our knowledge, no prior work has investigated planning-based approaches for legal text generation.

B Implementation Details

We implemented KeyBERT [1] with n-gram range of (1, 3). We enable Maximal Marginal Relevance (MMR) with a diversity value 0.7 to balance relevance and diversity. For TextRank (Mihalcea and Tarau, 2004), we utilize the implementation provided in the GitHub repository.

We perform LoRA fine-tuning with Mistral model for both our plan and response generator with an alpha and rank configuration in $\{8, 16\}$ and dropout of 0.1. We sweep over learning rates $\{1e-5, 3e-5, 5e-5, 1e-4, 3e-4\}$ and evaluating its performance on the validation set using 10K validation steps. We train the model end-to-end for one epoch using Adam Optimizer (Kingma and Ba, 2014). We perform inference over models with a temperature of 0.7 and nucleus sampling of 0.95.

[1] https://github.com/MaartenGr/KeyBERT

For retrieval, we dynamically populate the candidate datastore by excluding documents dated after the query case and including only those available up to the query date, thereby simulating a realistic setting. We segment these documents into paragraph chunks, to index them for retrieval. Prompt 1 and 2 detail the prompts used for content plan generation and response generations modules respectively.

C Case Study

Figure 3 presents a case study illustrating the impact of incorporating a keyphrase-based content planning mechanism as a forward-looking strategy for retrieving more relevant documents and guiding the generation process. The example pertains to an alleged violation of Article 6 §1 of the Convention, specifically addressing the unreasonable length of proceedings.

The first generation approach relies solely on the context to retrieve documents and generate a response. While this method produces a generally relevant next paragraph, it lacks forward-looking contextual alignment and tends to emphasize broad principles rather than addressing the specific circumstances of the case. In contrast, the keyphrase-based content planning mechanism employs a structured plan centered on critical themes such as "reasonableness of the length of proceedings," "complexity of the case," and "conduct of relevant authorities." This approach retrieves documents more pertinent to the next paragraph at hand, leading to tailored and contextually precise generation. By improving document selection and ensuring alignment with key case factors, this structured method enhances the relevance and effectiveness of the generated content, facilitating more robust legal reasoning.

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You are a judge at the European Court of Human Rights (ECHR), drafting a judicial judgment. The
paragraphs below reflect your reasoning thus far. Your task is to outline the next topics you intend
to address in the next paragraph of the judgment.

Instructions:
- Generate a maximum of 15 topics. Place higher-priority topics at the beginning of the list.
- Ensure the list is formatted as a comma-separated string.
- Ensure the topics align closely with the content of the previous paragraphs and the overall context
 of the judgment.

Input:
{{PREVIOUS PARAGRAPHS}}

Response:
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Prompt 1: Prompt for Keyphrase based content plan generation

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You are a judge at the European Court of Human Rights (ECHR), drafting a judicial judgment. Below are
 paragraphs you have already written as part of your reasoning. Your task is to write the next
paragraph to continue developing the judgment.

To assist you, excerpts from relevant external cases have been provided. These cases may offer
supporting arguments or precedents to strengthen your reasoning, but their use is optional. Your
primary focus should be on ensuring the logical flow and coherence of the judgment.

Instructions:

- Refer to external cases only if they add value to your reasoning or support the legal principles
being applied. When citing an external case, include a proper citation at the end. For example: (Case
 of X v. Y). If referencing a specific section or paragraph from an external case, denote it clearly
using the section symbol and the paragraph number.
- Ensure that the new paragraph is directly related to the content of the previous paragraphs and the
 broader context of the ECHR judgment.
- Use formal, precise, and structured language typical of ECHR judgments.


Input:
External Cases:
    {EXTERNAL CASES}

Previous Paragraphs:
    {PREVIOUS PARAGRAPHS}

Response:
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Prompt 2: Prompt for Response Generation Module

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Context:
THE FACTS
4. The applicants details and information relevant to the application are set out in the appended
table.
5. The applicant complained of the excessive length of enforcement proceedings that she had initiated
 on 3 August 2009 (her submission was delivered to the Humenn District Court on 4 August 2009).
..
..
THE LAW
ALLEGED VIOLATION OF ARTICLE 6-1 OF THE CONVENTION
..
..
..
16. It observes that the present application was communicated to the Government on 22 September 2019
and the applicant accepted the friendly settlement by a letter dated 15 October 2020 (delivered to
the Court on 23 October 2020), that is before the adoption of the second Constitutional Courts
decision. It was only after that date that the applicant learned about the award granted by the
Constitutional Court. Moreover, according to the domestic law, as long as the proceedings are pending
, the applicant is entitled to complain about their length before the Constitutional Court, including
 repeatedly. Accordingly, the Court is of the view that the applicant cannot be considered as having
failed to provide the Court with information concerning the very core of her case (see a contrario,
Hadrabov and Others v. Czech Republic (dec.), nos. 42165/02 and 466/03, 25 September 2007), and thus
dismisses the Governments objection in this respect.
17. As regards the applicants victim status, the Court notes that in its second decision the
Constitutional Court examined the overall length of the proceedings which had at that time lasted ten
 years and eight months (from 4 August 2009 until 7 April 2020) for one level of jurisdiction,
including the proceedings before the enforcement officer. It found that there had been a violation of
 the applicants right to a hearing without unjustified delay and awarded her EUR 3,000 EUR in just
satisfaction. Considering that the enforcement proceedings have been pending for more than eleven
years and pertain to the enforcement of non-pecuniary damages following the death of the applicants
husband, the Court finds that such an amount cannot be considered sufficient in the light of the
Courts case-law (see Scordino v. Italy (no. 1) [GC], no. 36813/97,  205-06 and 214-15, ECHR 2006V,).
The applicant can accordingly still claim to be a victim of a breach of the reasonable time
requirement.

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Next Paragraph Generation without planning:
Retrieved Documents:
 - Dobbertin v. France, 1993,  44;
 - Kart v. Turkey [GC], 2009,  68

Next paragraph:
18. The Court further emphasizes that the right to a hearing within a reasonable time, as enshrined
in Article 6  1 of the Convention, serves to protect individuals from prolonged uncertainty in legal
disputes and to uphold the effective administration of justice. In the present case, the excessive
duration of the enforcement proceedings, spanning over eleven years, represents a failure to adhere
to this fundamental guarantee.
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Next Paragraph Generation with keyphrase-based content planning:
Content Plan:
reasonableness of the length of proceedings, assessment of reasonableness, complexity of case,
conduct of the relevant authorities

Retrieved Documents:
 - Neumeister v. Austria, 1968,  21;
 - Pedersen and Baadsgaard v. Denmark, 2004,  45

Next Paragraph with planning mechanism:
18. The Court reiterates that the reasonableness of the length of proceedings must be assessed in the
 light of the specific circumstances of each case, taking into account criteria such as the
complexity of the matter, the conduct of the applicant and the relevant authorities, and the stakes
involved for the applicant (see Neumeister v. Austria, 1968,  21, and Pedersen and Baadsgaard v.
Denmark, 2004,  45). In the present case, the proceedings spanned over a decade, primarily due to
procedural delays and inefficiencies attributable to the authorities, including the enforcement
officers actions.
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Case Study 3: Comparison of generated next paragraphs using approaches with and without keyphrase-based content planning.